INTRODUCTION TO AEROSPACE

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Could Hydrogen fuel replace Kerosine in the near future? - Combustion in Air- and Spacecraft Propulsion

Even though kerosine engines have become more efficient over the years, they are not sustainable in the long term due to their high C02 emissions. On average, 3.16kg per 1kg of Kerosine [1]. One of the most promising alternative aviation fuels is hydrogen. Looking at the numbers, one can easily see why hydrogen is so promising. It carries three times the energy of conventional Kerosine, has zero C02 emissions and around 90% less nitrogen oxide emissions [2, page 3] . ,, When dealing with hydrogen as fuel, it is important to be aware that hydrogen is not a fuel in the sense of an energy source e.g., crude oil. In fact, it is an energy carrier, rather comparable to a battery. Hydrogen does not exist in pure state in nature but has to be separated under the expense of energy first, and only parts of this energy can be retrieved during its use afterwards,,[2]. According to the International Energy Agency (IEA), in 2021, the CO2 equivalent for the production of 1kg H2 was 12-13kg. However, their net zero by 2050 scenario includes a reduction of CO2 equivalent for hydrogen production to 6-7CO2-eq/kg H2 by 2030 and to <1kg CO2-eq/kg H2 by 2050 [3]. So, hydrogen as an alternative fuel is highly dependent on the evolution of hydrogen production. Additionally, the combustion of 1kg of hydrogen emits 9kg of water vapour, which is around 7 times more than the combustion of Kerosine. Under certain conditions, this can form contrails and cirrus clouds [2, page 3], which have a significant but not well-understood impact on climate change [6].

How does hydrogen fuel affect fuselage design?

To store the same energy, the LH2 tanks take up around 4 times more volume than Kerosine. Calculations on the weight of the LH2 tank structure have shown that wing tanks would be too heavy [4]. This and the additional need for pressurized tanks and cryogenic insulation makes LH2 fuel-less space efficient compared to conventional fuel tanks. All that comes at the cost of cargo or passenger space or an increase in fuselage size will result in higher drag. All this results in decreased efficiency. Thus, the operation costs for LH2-powered aircraft are about 2-3% higher than that of kerosine-powered aircraft. making H2 economical less appealing. [2, page 8] This value varies when comparing jet engines to turboprop engines, this article will focus on turbofan jet engines. Apart from that, other challenges like the effects of permeation, hydrogen embrittlement, and the possible build-up of ice on the aircraft have to be taken into account when designing LH2 Tanks. However, the efficiency of hydrogen engines can be increased by taking advantage of the cryogenic properties of liquid hydrogen [2].

Using Cryogenic Liquid Hydrogen in Unconventional Cycles.

The efficiency of the engine can be increased by utilizing the cryogenic properties of LH2 in an unconventional cycle. There are multiple possible cycles, which are discussed in detail in [5, page 27]. One promising cycle utilizes the Lh2 to cool some of the engine bleed air with a heat exchanger, this low-temperature bleed air is then used to cool the Low-pressure turbine blades. Due to better cooling, the turbine inlet temperature can be increased while maintaining the engine lifetime. This results in a higher power output by the low-pressure turbine, which in turn makes it possible to increase the bypass area, which makes the engine more efficient and powerful. Dries Verstraete references a paper by Boggia, S., Jackson, A., & Singh, R. (2001), "Unconventional cycles for aero gas turbine engines burning hydrogen" in which a V2427-A5 is used as an Example. Using the described unconventional cycle, the Bypass area was increased from 4.8 to 6.5,

therefore increasing the engine's thrust by 28% while keeping the Thrust-specific fuel consumption almost identical [5, page 27].

The effect of swept wings on the effective critical Mach number - Civil and Military Aircraft

An aeroplane creates lift by influencing the flow of air around the wing. One of the ways the wing produces lift is by creating a high-pressure area underneath the wing and a low-pressure area above the wing; this results in an increased airspeed above the wing.[7] As the plane approaches the speed of Mach one, the fast-moving air on top of the aerofoil can reach Mach one significantly earlier. This results in the formation of a shockwave on top of the aerofoil. The build-up of such a shockwave leads to airflow separation, as the air is no longer attached to the aerofoil; the aerofoil geometry does not affect the airflow and thus reduces lift and increases drag significantly. The speed at which this phenomenon occurs is called the critical Mach number and effectively limits the aircraft's top speed. It depends on the individual geometry of the aerofoil and is, therefore, different for every aircraft.[7] There are, however, different ways to overcome the limiting properties of the critical Mach number. One way to achieve this involves changing the sweep angle of the wings. An increase in the sweep angle separates the airflow above the wing into two major components. A spanwise component flows along the edge of the wing, and a chordwise component flows orthogonal to the wing. The chordwise flow is the flow component creating lift and is responsible for the formation of shockwaves[9]. So, an increase in the angle of attack results in a bigger spanwise component and, therefore, reduces the flow component in the chordwise direction and the velocity of the flow in the chordwise direction. Because the chordwise flow is responsible for the shockwave formation, the speed at which flow separation occurs is also dependent on the velocity of the chordwise flow[10][8]. This is called the effective Mach number: $M_{eff}=M\cos(\phi)$ (M_{eff} = effective Mach number, M = free stream Mach number, ϕ = sweep angle)[8]. This simplified explanation of the effects of the sweep angle on the effective Mach number, a more detailed explanation can be found in Jeffrey J. Kirkman's and Timothy T. Takahashi's article "Critical Mach Number Prediction on Swept Wings"[9] as it would exceed the scope of this text. The formula mentioned above shows that by increasing the sweep angle, aircraft like the F-16 can achieve an effective Mach number of around 1,5 Mach, well above Mach 1. A negative effect of increasing the sweep angle is that as the velocity of the chordwise flow component is decreased, the aerofoil can produce less lift, which is especially problematic for fighter aeroplanes as they need sufficient lift to perform high g manoeuvres. An attempt to increase the lift of highly swept wings is by using vortex lift. The vortex flow relevant to achieving vortex lift describes an airflow spiralling around a rotation axis in the direction of flow. This results in an increase in flow speed and, thus, according to Bernoulli's principle, a decrease in pressure. If this vortex is created on top of the aerofoil, the pressure difference between the low-pressure area of the vortex and the high-pressure area underneath the wing results in lift. To create a vortex, the aircraft has to create a high- and a low-pressure area, which is the result of this pressure difference. In aviation, vortices are usually avoided as they result in increased drag; therefore, aircraft that don't utilize vortex lift avoid sharp edges as they make it more likely for the flow to detach, resulting in the formation of a vortex. Military aircraft take advantage of this flow characteristic by utilizing sharp edges, for example, in the delta wings with sharp leading edges,

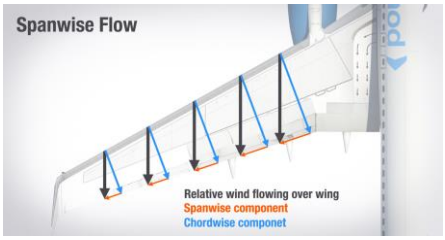


Fig. 1: Flow separation into spanwise and chordwise components

to achieve a stable vortex on top of the aerofoil. This effect increases with the angle of attack, resulting in a higher-pressure difference at the wing leading edge and, therefore, in a more powerful vortex. [11] This, in turn, results in high lift at high angles of attack, making it possible for the aircraft to perform high G manoeuvres.

Could there be life on the Jupiter moon Europa - Space Exploration

In the search for alien life, the Jupiter moon Europa is regarded as one of the most promising places in our Solar system. But why would the Jupiter moon, which is about one-fourth of Earth's radius and about 5.2 times further away from the Sun, be ideal for life? In the search for alien life, two things stand out as the most important: liquid water and energy. Europa is believed to have a subsurface ocean spanning the entire Moon. Due to the thick ice crust and far distance from the sun, it is unlikely that life on Europa could be dependent on photosynthesis. Characteristic cracks in this ice crust, however, indicate that Europa is being constantly deformed by Jupiter's gravity. This results in the Moon's core heating up, which in turn is going to heat up the subsurface ocean. This phenomenon could provide an alternative energy source for life to form around similar hydrothermal vents here on Earth.[12]

There is, to date, no undeniable evidence of a liquid ocean on Europa, but earlier missions and measurements show strong indications of its existence. The most compelling indication was found by the Galileo spacecraft as it measured Europa's magnetic field. It was discovered that Europa has an additional magnetic field, which is induced by Jupiter's magnetic field. For this phenomenon to occur, it is necessary for Europa to have a layer of electrically conducting liquid; this is a strong indication of a salty ocean possibly even saltier than Earth. Another strong indication is that Europa's surface is geologically young and has very few carter but very characteristic Surface features like Ridges, cycloids and linear cracks. These features are thought to be the result of tidal stresses, which in return require a liquid layer to create enough stress for these features to occur. The gravitational measurements from both Voyer and Galileo spacecraft show that the Moon is likely to have a metallic core with a radius of ~700km and a Layer of water around 150km thick. This layer of water is believed to be, at least partially, in a liquid state due to Europa's internal heat. This internal heat results from the Moon experiencing strong enough tidal forces from Jupiter to generate sufficient heat through internal friction[13]. This internal energy is Europa's most important energy source and plays a significant role in sustaining liquid water. This could result in a dynamic environment with a water cycle between the ocean floor and the ice crust and, therefore, provide the ocean layers with the oxidants and reductants that result from high radiation hitting the ice crust. These oxidants and reductants provide an additional energy source which could be utilized by life[14]. As the tidal forces flex the Moon's core, water could seep into the upper layer and get mixed with organic compounds and minerals, or it could result in hydrothermal regions[12][14]. These would be ideal conditions for life; however, little is known about the composition of the Europa oceans, and a lack of necessary elements could hinder the development. To date, the existence of an ocean on Europa, although it is very likely, is not confirmed. In order to learn more about the possibility of life on Europa we need to send more missions to Europa. One of these Missions is NASA's Europa Clipper.

The Europa Clipper spacecraft will launch on the 14th of October, 2024, and will reach Jupiter orbit in April 2030. It will further map the Ice crust for elevation and thickness and study the composition of the atmosphere and the ice crust using its two spectrometers (EUROPA-UVS, MISE). It will further analyze Europa's magnetic field to gather more information about how the

subsurface ocean must behave. It will map Europa's deformation in regard to Jupiter's tidal forces in order to better understand Europa's structure. It will also study how Jupiter's intense radiation impacts the ice crust and how it alters compounds. It will also carry a dust analyzer, which will collect and analyze any substances that might have been broken off or vented into space[15].

Project Breakthrough Starshot – Human Spaceflight Technologie

Project Breakthrough Starshot is trying to create a proof of concept for a solar sail-powered probe that can reach the system Proxima Centauri and observe the earth-like planet Proxima Centauri b within a human lifespan [17]. Robert Forward first proposed the principle of a spacecraft being powered by a solar sail in 1962 [17]. He proposed that by creating a very light probe, attaching it to a highly reflective sail, and shooting it with a powerful laser, one could harness the impulse of a photon colliding with the sail. The fact that such a spacecraft would not have to carry its fuel makes it possible to achieve previously unreachable speeds, with Starshot aiming for 0.2 times the speed of light[16].

For such a lightsail to work, it must convert as much of the light's impulse into kinetic energy as possible. To achieve that, it has to be made out of a very thin and light material with good reflective properties to reflect a large amount of photons and, therefore, achieve the most acceleration[16][17][19]. This can be achieved by maximizing the quotient of the Reflectance divided by the sail thickness[17]. Another way to increase acceleration is by increasing the power of the propelling beamer. This increase is, however, limited by the solarsail because it absorbs a small amount of the photons, which causes it to heat up. Therefore, the sail must have good radiative properties to radiate much of this heat into space[17][18]. It must also be durable enough to survive a high acceleration of up to 14900G and a journey of around 20 years[17]. As the Probe accelerates to one-fourth the speed of light, it starts to experience a relativist redshift in the accelerating beam, meaning the Solarsail has to have good reflective properties across a range of different wavelengths. Especially in the near-infrared Spectrum, as this wavelength is good at penetrating the Earth's atmosphere and is therefore suitable for an Earth-based beamer[19]. Suitable materials are Silicon and Diamont, which have good reflective properties and absorb only very small amounts of photons[17][19]. Those properties could theoretically be drastically enhanced by using them in photonic crystal slabs. This topic is discussed in detail in ,,Materials challenges for the Starshot lightsail,, reference [19]. As the sail is supposed to have an area of around $10m^2$ and be about as thin as hair[17][19], the slabs would have to be manufactured as individual tiles and connected with an Aerogel mesh. Which is beneficial as Aerogels have very good radiative properties.[19]

The earth-stationed beamer, as of 2018, was envisioned to have a power output in the primary optic of around $37kW/m^2$, which would accelerate the Starshot probe at 2500-14900G to 0.2 times the speed of light in 9 minutes with an increase in power output after 3 minutes to compensate for photon spillage around the sail due to its increased distance[2]. The Probe will ,,ride,, the laser beam, meaning the beam will be directed at the target location, and the Probe will adjust its orientation either actively or passively while being accelerated, as it would be near impossible to adjust the laser position relative to the Probe once it gets close to its target speed.[2][1]

The actual Probes themselves, called StarChip, will weigh less than 1 kg. In a paper from MIT, Andrew Sorenson suggests the use of superconducting wires on the spacecraft as they are highly energy efficient and have no need for additional heating because of their very low operating

temperatures. This minimizes the need for electricity and would enable the Probe to have a much smaller power unit. However, further research is needed to determine if the technology is applicable in this environment as Sorenson highlights the fact that superconductors are prone to radiation and, due to the lightweight of the StarChip probe, it will most likely not carry sufficient radiation shielding. This will increase the chance of system failure. Sorenson's hypothesis is that this probability can be drastically reduced by implementing shunted interconnections, which, in essence, are backup connections to increase System durability. [3]

Conclusion: solarsails are a promising technology that can open the door to interstellar exploration; however, many challenges remain to overcome.

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